Gradient Feature-Oriented 3-D Domain Adaptation for Hyperspectral Image Classification

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***Abstract*— Domain adaptation, which cleverly applies the classifier learned from the source domain with sufficient labeled samples to the target domain with limited labeled samples, provides a feasible alternative to handle the small training sample problem of hyperspectral image (HSI) classification and has attracted much attention in the research field recently. Apparently, feature discriminative ability is vital for domain adaptation, which plays a crucial role during the migration process of transfer learning. In this article, a gradient feature-oriented 3-D domain adaptation (GF-3DDA) approach is proposed for HSI classification. First, 3-D Gabor is employed to remove noise from the original data, and two 2-D gradient-based features, 2-D Sobel gradient (SG) and 2-D derivative-of-Gaussian (DtG), are extended to the 3-D domain to coincide with the integrated spatial–spectral organization of HSI. Thus, the 3-D Sobel–Gabor gradient (3DSGG) and 3-D derivative-of-Gaussian-Gabor (3DDGG) features are achieved. Second, a 3-D domain adaptation method is implemented to jointly exploit the second- and fourth-order statistical descriptors in the spatial–spectral dimensions, which could effectively reduce domain shifts and thus achieve improved domain adaptation. Third, all the extracted domain-adapted feature modules are collaboratively classified by extreme learning machine (ELM), and the probability-like outputs of every ELM classifier are combined together to accomplish the classification task. Four hyperspectral data sets that each contains two scenes, i.e., Pavia, Shanghai–Hangzhou, Indiana, and Houston, are tested in the experiments. When only ten labeled samples per class are used in the target domain, the classification accuracies on four hyperspectral data sets achieved by our GF-3DDA approach are 93.31%, 84.35%, 69.32%, and 80.06%, respectively.**

***Index Terms*— Domain adaptation, feature extraction, hyperspectral image (HSI) classification.**

I. Introduction

With the development of sensor technology, applications of hyperspectral image (HSI) have been largely expanded, including agriculture, geology, and environmental monitoring [1], [2]. HSI classification belongs to the pattern recognition field, which predicts a single label to each spatial pixel. Unlike multispectral image that only possesses dozens of channels, HSI often contains hundreds of spectral bands, which provides more possibility to identify surface materials. Unfortunately, the large spectral redundancy in HSI also increases the difficulty of classification in turn [3], [4]. Correspondingly, feature extraction [5], [6] and band selection [7]–[10] methods have been extensively exploited in HSI processing. For instance, Kang *et al.* [11], [12] proposed a HSIs feature extraction method based on image fusion and intrinsic image segmentation, respectively, which proved that feature extraction can effectively improve the accuracy of HSI classification.

Another challenge is the scarce labeled samples in the scene (hereafter referred to as target domain) since manually labeling is very hard and needs much effort in practice. Fortunately, it is meaningful and valuable to consider a circumstance that when we have sufficient labeled samples in another scene (referred to as source domain) and both scenes are related to each other in some aspects (such as two cities containing road, buildings, plants, and so on); the classification knowledge obtained from the source domain can be adapted to the target domain by transferring knowledge [13]. In general, domain adaptation can be divided into two categories according to the different characteristics of the source domain and target domain, i.e., homogeneous and heterogeneous domain adaptation. For instance, Qin *et al.* [14] proposed a homogeneous domain adaptation method based on tensor alignment of HSI. Both the source domain and target domain were first segmented into superpixels, and the tensors of the two domains were constructed to

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include adjacent samples of a single pixel. Then, the original tensor was projected to the invariant quantum space as the core tensor by applying the Tucker decomposition method. In heterogeneous domain adaptation, the feature spaces of the source domain and target domain are inconsistent or feature representations are different. Wang *et al.* [15] proposed an unsupervised heterogeneous domain adaptation method for HSI classification, which captures convertible features with a small amount of similar data. Liu *et al.* [16] also proposed an unsupervised heterogeneous domain adaptation method that uses a knowledge transfer principle and an angle-based master metric method to measure the distance between the source and target domains. There are many other methods to adapt to homogeneous [17] and heterogeneous domains [18]–[20] in the literature.

Concerning HSI classification, the data distribution of the source and target scenes is different in most cases [13], [21]. If the classification model trained in the source domain is directly applied to the target domain, it may cause the situation called negative transfer that could degrade the classification performance rather than improving it. This phenomenon is technically known as domain shifts [22], which is caused by the difference of data distribution, spectral band, ground sample distance, and other factors [23]. Fortunately, we can solve this problem by applying domain adaptation algorithms without tagging new data [24]–[26]. Specifically, domain adaptation mainly reduces domain shifts by aligning the relevant information from the source and target domains [27], which can be further categorized into classifier- and feature-level domain adaptation. In classifier-level domain adaptation [28], the goal is to find an appropriate classifier so that the model trained in the source domain can roughly predict the test data in the target domain. A classifier-level method called domain transfer ranked support vector machines (DTRSVMs) [29] is proposed for person reidentification problem. Ma *et al.* [30] proposed a classifier-level domain adaptation method by using adaptive ranking support vector machines (SVMs) to learn distance models. A discriminative model with low weight in target negative image pairs and high weight in target positive mean is trained by relaxing the constraint to a necessary condition.

On the other hand, feature-level domain adaptation methods reduce domain changes through building feature subspaces that could represent common features of the source and target domains [31]–[33] or directly align source and target domain data [34]. Ye *et al.* [35] proposed a domain adaptation method based on dictionary learning in feature level, which tried to find a structure that could describe the data eigenspace in a linear combination of basis atoms. Remus [36] utilized the Jansen–Shannon divergence and domain homogeneity to measure the similarity and complexity between source and target domains, respectively. Pengfei *et al.* [37] proposed a domain-specific feature migration method for different scene transfer learning. It exploits the relationships of different domains by looking for the specific characteristics of the domain itself and uses the maximum mean discrepancy metric to minimize the distribution contradiction in the field. Sun *et al.* [38] proposed a hyperspectral classification method based on extended random walker, which uses the spatial correlation between the adjacent pixels and the connectivity between training samples and test samples for modeling and makes full use of the high spatial–spectral information of the image. In addition, Sun *et al.* [34] proposed a correlation alignment (CORAL) technique that aligns the feature distribution of the source and target domains by employing the second-order statistics of the feature. However, CORAL has the disadvantage that the aligned dimension of the source and target domains should be the same, and thus, the generalization of CORAL is limited. Therefore, it is desirable to overcome the limitation and improve the practicability of the CORAL method.

In particular, it should be specially emphasized that except for the major role of domain adaptation in transfer learning, the discriminative ability of features (in both source and target domains) is also very important for land cover classification. In this article, a gradient feature-oriented 3-D domain adaptation (GF-3DDA) approach is proposed for HSI classification. Specifically, 3-D Gabor [39] filtering was first used to remove noise and interference information. Then, feature extraction methods based on 3-D Sobel–Gabor gradient (3DSGG) and 3-D derivative-of-Gaussian-Gabor (3DDGG), which are, respectively, the extensions of 2-D Sobel gradient (SG) [40] and 2-D derivative-of-Gaussian (DtG) [41], are developed to make full use of the integrated spatial–spectral information of HSI. It is worth mentioning that although the 3-D derivative of Gaussian transform has been proposed in [42], which used different orders of Gaussian transform, our proposed 3DDGG model only employs the second-order Gaussian derivative to make a tradeoff between model representability and complexity.

With regard to the domain adaptation procedure, the CORAL method is first enhanced to break the restriction of requiring the aligned dimension of source and target domain data to be the same, and then, the boosted CORAL (BCORAL) method is performed on both spectral and spatial dimensions, called 3-D boosted CORAL (3DBCORAL), to further improve the discriminative ability of attained features. Meanwhile, in addition to second-order statistics [43], the fourth-order statistics that characterizes the non-Gaussianity of data distribution is an important supplement and imported to enhance domain adaptation. Hence, the 3DBCORAL operator with two statistical quantities is referred to as 3DBCORAL-SF. After applying 3DBCORAL-SF domain adaptation on the extracted 3DSGG and 3DDGG features, a number of domain-adapted feature data, including different orientations and statistical measures, is achieved. Finally, extreme learning machine (ELM) [44] classifier is selected to train the classification model on each domain-adapted feature module sequentially, and the probability-like outputs of every ELM classifier are associated together in a multitask scheme. The flowchart of the proposed GF-3DDA framework is shown in Fig. 1. The main contributions of the proposed GF-3DDA approach are listed as follows.

1) The 3-D Gabor wavelet is used to filter the HSI to remove noise and interference information. We then extend the 2-D SG and 2-D second-order Gaussian derivative into the 3-D domain to coincide with the

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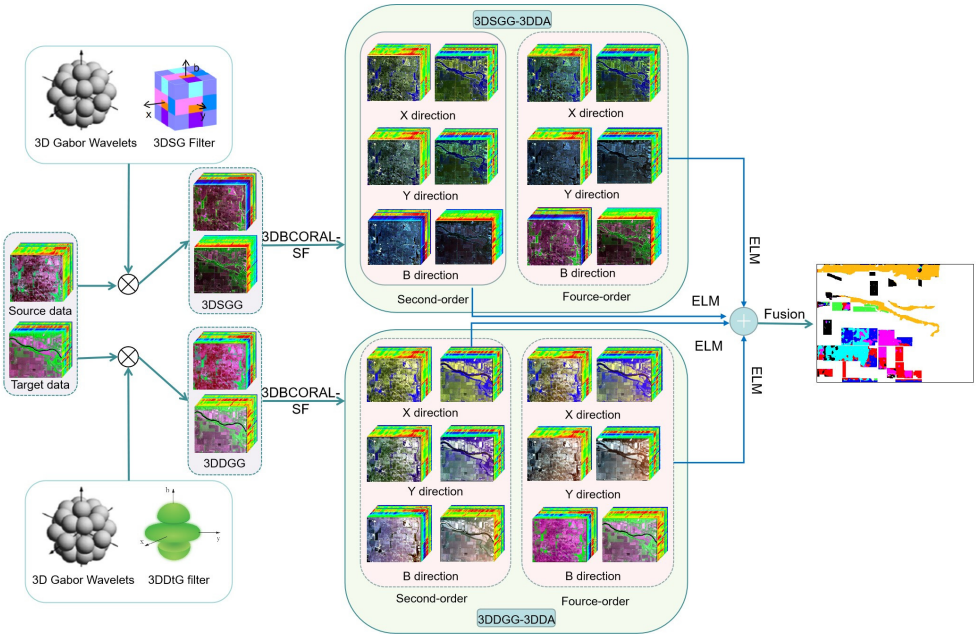


Fig. 1. System diagram of the proposed GF-3DDA framework for HSI classification.

integrated spatial–spectral organization of HSIs, which are, respectively, named 3DSGG and 3DDGG. Specifically, 3DSGG can effectively represent the geometry structure of land covers, whereas 3DDGG can sufficiently characterize the texture information of the objects (only the second-order Gaussian derivative is employed to make a tradeoff between model representability and complexity). Apparently, both the 3DSGG and 3DDGG features reveal the internal structure of an HSI from different aspects, and hence, they should be jointly concerned in a unified framework.

2) The boosted 3DCORAL technique breaks the limitations of traditional CORAL that requires the aligned dimension of the source and target domain to be the same, and thus, the universality and practicability of BCORAL is greatly enhanced. Meanwhile, since 3DBCORAL is employed in the spectral and two spatial directions, the discriminative information inside HSI is fully exploited. Furthermore, the two statistical descriptions (second order and fourth order) address the data properties from different aspects, and therefore, they can provide complementary feature representation and the discriminative ability of achieved feature data can be improved.
3) Instead of simply concatenating all the extracted domain-adapted feature cubes, the ELM classifier, which has several merits in data processing (such as high robustness and low complexity), is applied on a dozen extracted feature modules in turn, and the results are then associated in a multitask learning scheme, which not only takes full advantage of feature discriminative ability of each data cube (the classification performance is guaranteed) but also greatly decreases the time complexity of GF-3DDA (the dimension of concatenated feature is huge).
4) All the parameters contained in our GF-3DDA approach can be either preset and kept unchanged for multiple experimental data sets or learned during the training process; therefore, the generalization ability of the method can be ensured. Meanwhile, the 3-D feature extraction and domain adaptation procedure are obviously unsupervised and unrelated to the training set; therefore, they need to be computed only once and can be accomplished in advance, which further improves the efficiency of the proposed GF-3DDA approach.

Eight recent and relevant state-of-the-art methods are conducted to compare with the proposed GF-3DDA, including two feature extraction methods (extended multiattribute profiles (EMAPs) [45] and local binary pattern (LBP) [46]), four domain adaptation methods (transfer component analysis (TCA) [47], geodesic flow kernel (GFK) [48], selective pseudo-labeling (SPL) [49], and TwoCnn [50]), and two deep learning classification methods (convolutional neural network (CNN) [51] and residual network (ResNet) [52]). Since domain adaptation methods are mainly concerned in this article, each data set contains two real HSIs, including Pavia, Shanghai–Hangzhou, Indiana, and Houston. Experimental results constantly demonstrate the superiority of GF-3DDA over the compared ones.

The rest of this article is organized as follows. Section II introduces the related works, including the two 2-D feature extraction methods, 3-D Gabor wavelets, domain adaptation, and ELM. Next, Section III presents the proposed GF-3DDA approach for HSI classification. Experimental setup and ablation analysis are provided in Section IV, and detailed

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experimental results are given in Section V. Finally, the conclusion is given in Section VI.

II. Related Works

A. 2-D Sobel-Based Gradient

The 2-D Sobel filter is a popular operator to extract the gradient information of an image, which is efficient of describing the geometric structure of the image [53], [54]. The 2-D SG feature map not only highlights the edge details but also captures local optical information. In practice, for a 2-D image $\mathbf{I} \in \mathbb{R}^{X\times Y}$, the horizontal and vertical Sobel templates $\mathbf{S}_x \in \mathbb{R}^{3\times 3}$ and $\mathbf{S}_y \in \mathbb{R}^{3\times 3}$ are, respectively, convolved with $\mathbf{I}$ to measure the horizontal and vertical gradient

$$\mathbf{I}_x = \mathbf{S}_x \otimes \mathbf{I} \tag{1}$$

$$\mathbf{I}_y = \mathbf{S}_y \otimes \mathbf{I} \tag{2}$$

$$\mathbf{S}_x = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \quad \mathbf{S}_y = \mathbf{S}_x^T \tag{3}$$

where $\otimes$ and $(\cdot)^T$ are, respectively, the convolution and transformation operator. After obtaining the horizontal and vertical gradient $\mathbf{I}_x$ and $\mathbf{I}_y$, the 2-D SG feature map $\mathbf{M} \in \mathbb{R}^{X\times Y}$ is computed as follows:

$$\mathbf{M} = \sqrt{\mathbf{I}_x^2 + \mathbf{I}_y^2}. \tag{4}$$

B. 2-D Derivative-of-Gaussian

DtG [41] is an important tool in image feature extraction, and the features extracted by DtG can effectively represent the texture information of the image. So far, most DtG feature extraction methods are based on one or two dimensions. In order to better understand the 2-D DtG method, let us consider the 1-D Gaussian function $G(x;\sigma)$ and its derivative first

$$G(x;\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \tag{5}$$

where σ is the scale parameter. The nth-order derivative $\mathcal{D}^{(n)}(x;\sigma)$ of $G(x;\sigma)$ is

$$\mathcal{D}^{(n)}(x;\sigma) = \frac{\partial^n}{\partial x^n} G(x;\sigma), \quad n = 1, 2, 3, \ldots \tag{6}$$

After a number of deduction, (6) can be written as

$$\begin{aligned}\mathcal{D}^{(n)}(x;\sigma) &= \frac{1}{\sigma\sqrt{2\pi}} \frac{\partial^n G(x;\sigma)}{\partial x^n} \exp\left(-\frac{x^2}{2\sigma^2}\right) \\ &= \frac{(-1)^n}{(\sigma\sqrt{2})^n} H_n\left(\frac{x}{\sigma\sqrt{2}}\right) G(x;\sigma), \quad n = 1, 2, 3, \ldots\end{aligned} \tag{7}$$

where H_n is the nth Hermite polynomials. Likewise, a 2-D Gaussian function $G(x, y;\sigma)$ is defined as

$$G(x, y;\sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{8}$$

and the derivative $\mathcal{D}^{(n)}(x, y;\sigma)$ is

$$\mathcal{D}^{(n)}(x, y;\sigma) = \frac{\partial^n}{\partial x^n} G(x, y;\sigma) + \frac{\partial^n}{\partial y^n} G(x, y;\sigma) \quad n = 1, 2, 3, \ldots \tag{9}$$

In practice, the 2-D DtG filter is obtained by discretizing $\mathcal{D}^{(n)}(x, y;\sigma)$ in a predefined area. Similar to the filter size of 2-D SG, the filter size of 2-D DtG is also 3×3, that is, $\mathcal{D}^{(n)}(x, y;\sigma) \in \mathbb{R}^{3\times 3}$. Regarding a 2-D image $\mathbf{I} \in \mathbb{R}^{X\times Y}$, the nth 2-D DtG feature map $\mathbf{G}^{(n)}(x, y;\sigma) \in \mathbb{R}^{X\times Y}$ can be thus obtained by

$$\mathbf{G}^{(n)}(x, y;\sigma) = \mathcal{D}^{(n)}(x, y;\sigma) \otimes \mathbf{I}. \tag{10}$$

C. 3-D Gabor Wavelets

Gabor wavelet is a useful tool to characterize the joint time and frequency relationship of collected signal [55]. In the 3-D Gabor wavelet, the response size of each sampling position (x, y) of the spectrum band b contains rich information about the signal change in local area. In particular, the circular 3-D Gabor wavelet is defined as follows:

$$\Phi_{f,\theta,\varphi}(x, y, b) = \frac{1}{s} \times \exp(j2\pi(ux, \upsilon y, \omega b)) \times \exp\left(-\frac{x^2+y^2+b^2}{2\hat{\sigma}}\right) \tag{11}$$

where

$$\begin{aligned} s &= (2\pi)^{(3/2)}\hat{\sigma}^3 \\ u &= f\sin\varphi\cos\theta \\ \upsilon &= f\sin\varphi\cos\theta \\ \omega &= f\cos\varphi. \end{aligned}$$

Among them, (x, y, b) represents the position of the hyperspectral pixel, s is the normalized value, and $\hat{\sigma}$ is the width of the Gaussian envelope in the spatial–spectral domain. In addition, φ and θ are the angles of the wave vector in the frequency domain, and f is the center frequency of the sin plane wave. In our experiment, only the Gabor filter that is parallel to the spectral orientation is reserved, and the amplitude of Gabor wavelets with four different scales is used [56].

D. Correlation Alignment

The main challenge in transfer learning is the domain shift between the source domain and target domain, and domain adaptation is a representative method in transfer learning. Compared with supervised domain adaptation methods, unsupervised domain adaptation is preferred when dealing with the problem of lacking labels.

CORAL is an unsupervised domain adaptation method proposed by Sun *et al.* [34]. In this approach, domain changes are minimized by aligning the second-order statistics of the source and target domains. Given source domain data $\mathbf{D}_S \in \mathbb{R}^{U\times Z}$ and target domain data $\mathbf{D}_T \in \mathbb{R}^{W\times Z}$ (note that the aligned dimension of $\mathbf{D}_S$ and $\mathbf{D}_T$ should be the same), the covariances

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$\mathbf{C}_S \in \mathbb{R}^{Z\times Z}$ and $\mathbf{C}_T \in \mathbb{R}^{Z\times Z}$ of the source and target data can be calculated by

$$\mathbf{C}_S = \mathrm{Cov}(\mathbf{D}_S) + \mathbf{E} \tag{12}$$

$$\mathbf{C}_T = \mathrm{Cov}(\mathbf{D}_T) + \mathbf{E} \tag{13}$$

where $\mathrm{Cov}(\cdot)$ denotes computation of covariance. The added identity matrix $\mathbf{E} \in \mathbb{R}^{Z\times Z}$ is used to avoid singularity of the covariance. Finally, the aligned source $\overline{\mathbf{D}_S} \in \mathbb{R}^{U\times Z}$ and target domain data $\overline{\mathbf{D}_T} \in \mathbb{R}^{W\times Z}$ are computed by

$$\overline{\mathbf{D}_S} = \mathbf{D}_S \mathbf{C}_S^{-\frac{1}{2}} \mathbf{C}_T^{\frac{1}{2}} \tag{14}$$

$$\overline{\mathbf{D}_T} = \mathbf{D}_T \mathbf{C}_T^{-\frac{1}{2}} \mathbf{C}_S^{\frac{1}{2}}. \tag{15}$$

E. Extreme Learning Machine

ELM is a structure using a one-way single-layer feedforward neural (SLFN) network [44], [57]. Compared with SVM and sparse logistic regression (SLR), ELM is advantageous in fast training speed and noise-resistant robustness. Suppose that $\mathbf{A} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{N}$ represents the N training samples, where $\mathbf{x}_i \in \mathbb{R}^B$ is the input feature vector (B is the feature number) and $\mathbf{y}_i \in \mathbb{R}^C$ is the corresponding one-hot encoding label vector (C is the class number)

$$\mathbf{y}_i(c) = \begin{cases} 1 & \mathbf{y}_i \in \text{Class } c \\ 0 & \text{otherwise} \end{cases} \quad c = 1, \ldots, C. \tag{16}$$

For an SLFN network with K hidden nodes, the objective function of the ELM is defined as

$$E = \sum_{j=1}^{N} \left(\sum_{k=1}^{K} \beta_k f(\mathbf{W}_k \cdot \mathbf{x}_j + \mathbf{b}_k) - \mathbf{y}_j \right)^2 \tag{17}$$

where β_k is the output weight and $f(\cdot)$ is the activation function that can be any nonlinear piecewise continuous function (sigmoid function is used commonly). $\mathbf{W}_k$ is the input weight vector connecting the hidden nodes and the input nodes, and $\mathbf{b}_k$ is the bias of the kth hidden node. After the objective function of (17) has been optimized, the predicted label of a test sample $\mathbf{z}$ is determined by

$$\mathcal{L}_{\mathbf{z}} = \underset{c=1,\ldots,C}{\arg\max} \sum_{k=1}^{K} \beta_k f(\mathbf{W}_k \cdot \mathbf{z} + \mathbf{b}_k). \tag{18}$$

III. Proposed GF-3DDA Approach

In this section, two gradient-based feature extraction methods, 3DSGG and 3DDGG, for HSI are first presented. In the following, the 3-D BCORAL method with statistics fusion (3DBCORAL-SF) is proposed. Finally, the ELM-oriented multitask learning scheme is given. Some mathematical symbols used are listed in Table I. Before extracting 3DSGG and 3DDGG features, Gabor wavelet filtering is performed on both source $\mathbf{R}_S \in \mathbb{R}^{X\times Y\times B_{S'}}$ and target $\mathbf{R}_T \in \mathbb{R}^{M\times P\times B_{T'}}$ HSIs, and the corresponding filtered Gabor amplitude features are, respectively, $\mathbf{R}_{\mathrm{GS}} \in \mathbb{R}^{X\times Y\times B_S}$ and $\mathbf{R}_{\mathrm{GT}} \in \mathbb{R}^{M\times P\times B_T}$.

TABLE I
Definitions of Mathematical Symbols Used in Section III

Abbreviation	Meaning
$\mathbf{R}_S, \mathbf{R}_T$	Source and target domain data
$\mathbf{R}_{GS}, \mathbf{R}_{GT}$	Gabor feature of source and target domain
$\mathbf{V}_S, \mathbf{V}_T$	3DSGG feature of source and target domain
$\mathbf{G}_S, \mathbf{G}_T$	3DDGG feature of source and target domain
$\overline{\mathcal{V}_S^{(i)}}, \overline{\mathcal{V}_T^{(i)}}, i = 1, 2, 3$	aligned 3DSGG feature using second-order statistics
$\hat{\mathcal{V}_S^{(i)}}, \hat{\mathcal{V}_T^{(i)}}, i = 1, 2, 3$	aligned 3DSGG feature using fourth-order statistics
$\overline{\mathcal{G}_S^{(i)}}, \overline{\mathcal{G}_T^{(i)}}, i = 1, 2, 3$	aligned 3DDGG feature using second-order statistics
$\hat{\mathcal{G}_S^{(i)}}, \hat{\mathcal{G}_T^{(i)}}, i = 1, 2, 3$	aligned 3DDGG feature using fourth-order statistics

A. 3-D Sobel–Gabor Gradient

In order to fully exploit the rich spatial–spectral information in HSI, 2-D SG is extended to 3-D version, which is called 3-D Sobel gradient (3DSG). Similar to the 2-D Sobel filters defined in (3), the spatial-oriented 3-D Sobel filters, $\mathcal{S}_x$ and $\mathcal{S}_y$, are defined as follows:

$$\mathcal{S}_x = \mathrm{ReM}(\mathbf{S}_x, [1, 1, 3]) \tag{19}$$

$$\mathcal{S}_y = \mathrm{ReM}(\mathbf{S}_y, [1, 1, 3]) \tag{20}$$

where the ReM operator repeats the copies of array in the designated dimension. For the spectral-oriented Sobel filters $\mathcal{S}_b$, it is specially defined as

$$\begin{aligned} \mathcal{S}_b(:,:,1) &= \begin{bmatrix} -1 & -2 & -1 \\ -1 & -2 & -1 \\ -1 & -2 & -1 \end{bmatrix} \\ \mathcal{S}_b(:,:,2) &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \mathcal{S}_b(:,:,3) &= \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}. \end{aligned} \tag{21}$$

For the Gabor feature cube in the source domain $\mathbf{R}_{\mathrm{GS}}$, the gradient of corresponding direction is computed by

$$\mathbf{\Omega}_i = \mathcal{S}_i \otimes \mathbf{R}_{\mathrm{GS}}, \quad i \in \{x, y, b\}. \tag{22}$$

Meanwhile, since the pixels in neighborhood belong to the same class with a high probability, the Gabor feature $\mathbf{R}_{\mathrm{GS}}$ convolved with a 3-D mean filter is also considered to enhance the descriptiveness of the obtained feature [58]. Eventually, the gradient magnitude is combined with the mean filtered data, and the resultant 3DSGG feature cube $\mathbf{V}_S \in \mathbb{R}^{X\times Y\times B_S}$ is

$$\mathbf{V}_S = \sqrt{{\Omega_x}^2 + {\Omega_y}^2 + {\Omega_b}^2} + \mathrm{ReM}\left(\frac{1}{27}, [3, 3, 3]\right) \otimes \mathbf{R}_{\mathrm{GS}}. \tag{23}$$

Likewise, for the Gabor feature cube in the target domain $\mathbf{R}_{\mathrm{GT}} \in \mathbb{R}^{M\times P\times B_T}$, the corresponding 3DSGG feature cube $\mathbf{V}_T \in \mathbb{R}^{M\times P\times B_T}$ can be attained in the same way.

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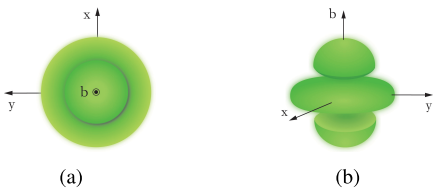


Fig. 2. 3DDtG filter. (a) Top view. (b) Side view.

B. 3-D Derivative-of-Gaussian-Gabor

As aforementioned, it is valuable to extend the 2-D DtG model to 3-D domain to make full use of the inherent texture information of HSI. The 3-D Gaussian function is developed based on the definition of 2-D Gaussian function in (8)

$$G(x, y, b; \sigma) = \frac{1}{(2\pi)^{3/2}\sigma^3} \exp\left(-\frac{x^2 + y^2 + b^2}{2\sigma^2}\right). \quad (24)$$

The corresponding nth derivative is calculated as

$$\mathcal{D}^{(n)}(x, y, b; \sigma) = \frac{\partial^{(n)} G}{\partial x^{(n)}} + \frac{\partial^{(n)} G}{\partial y^{(n)}} + \frac{\partial^{(n)} G}{\partial b^{(n)}}, \quad n = 1, 2, 3, \ldots \quad (25)$$

In our proposed 3DDGG model, the second Gaussian derivative ($n = 2$) is chosen to make a tradeoff between model representative ability and complexity. Then, each part in (25) can be deduced and expressed as

$$\frac{\partial^{(2)} G}{\partial i^{(2)}} = G(x, y, b; \sigma) \frac{(i^2 - \sigma^2)}{\sigma^4}, \quad i \in \{x, y, b\}. \quad (26)$$

Fig. 2 illustrates the sketch map of 3DDtG filter. Similarly, the 3DDtG filter is discretized for convolution, which has the size of $3 \times 3 \times 3$. For the Gabor feature cube in the source domain $\mathbf{R}_{\mathrm{GS}}$, the corresponding 3DDGG feature cube $\mathbf{G}_S \in \mathbb{R}^{X \times Y \times B_S}$ is obtained by

$$\mathbf{G}_S = \mathcal{D}^{(2)}(x, y, b; \sigma) \otimes \mathbf{R}_{\mathrm{GS}}. \quad (27)$$

For the Gabor feature cube in the target domain $\mathbf{R}_{\mathrm{GT}} \in \mathbb{R}^{M \times P \times B_T}$, the corresponding 3DDGG feature cube $\mathbf{G}_T \in \mathbb{R}^{M \times P \times B_T}$ can be attained in the same way.

C. 3-D BCORAL With Statistics Fusion

Generally, domain adaptation of CORAL is only performed in the spectral dimension. Obviously, the correlation of HSI exists not only in the spectral dimension but also in the spatial dimension. However, the CORAL method requires that the aligned dimension of both source and target domain data must be the same, whereas the spatial size of the two data sets is different from each other in most cases. To break the dimension-consistent limitation of CORAL and enrich the representative power of extracted features, a 3DBCORAL method is proposed in this article to align the source and target data in three orientations of HSI (it should be emphasized that our 3DBCORAL method can also handle the circumstance that the spectral bands of the two data are different). Alternatively, since different statistical measures describe the characteristics of HSI from different aspects, the fourth-order statistics reflecting the non-Gaussianity of data distribution is concerned in addition to traditional-used second-order statistics. Thus, 3DBCORAL with statistics fusion is named 3DBCORAL-SF.

Here, we take the 3DDGG feature, $\mathbf{G}_S \in \mathbb{R}^{X \times Y \times B_S}$ and $\mathbf{G}_T \in \mathbb{R}^{M \times P \times B_T}$ of source domain and target domain, as an example to show the process of 3DBCORAL-SF, and the horizontal alignment in spatial domain (the x-axis) is considered (the other directions can be handled in the same way). First, the horizontal dimension of $\mathbf{G}_T$, i.e., M, is scaled down to X by nearest neighbor interpolation algorithm (suppose that $M > X$ and vice versa), and the scaled 3DDGG in target domain $\mathbf{G}_{T_S} \in \mathbb{R}^{X \times P \times B_T}$ is achieved. Second, in order to facilitate the calculation of second- and fourth-order statistics, three 3DDGG feature cubes, including $\mathbf{G}_S$, $\mathbf{G}_T$, and $\mathbf{G}_{T_S}$, are stretched along the direction to be aligned. Concretely, three stretched 2-D feature data are, respectively, denoted as $\mathbf{G}_S^{(1)} \in \mathbb{R}^{X \times (Y \times B_S)}$, $\mathbf{G}_T^{(1)} \in \mathbb{R}^{M \times (P \times B_T)}$, and $\mathbf{G}_{T_S}^{(1)} \in \mathbb{R}^{X \times (P \times B_T)}$. Since alignment is applied on $\mathbf{G}_S^{(1)}$ and $\mathbf{G}_{T_S}^{(1)}$, the corresponding fourth-order statistics, $\mathbf{K}_S \in \mathbb{R}^{X \times X}$ and $\mathbf{K}_{T_S} \in \mathbb{R}^{X \times X}$, can be represented by

$$\mathbf{K}_S = \mathrm{Cok}\left(\mathbf{G}_S^{(1)}\right) + \mathbf{E} \quad (28)$$

$$\mathbf{K}_{T_S} = \mathrm{Cok}\left(\mathbf{G}_{T_S}^{(1)}\right) + \mathbf{E}. \quad (29)$$

$\mathrm{Cok}(\cdot)$ denotes cokurtosis and is calculated as

$$\mathrm{Cok}\left(\mathbf{G}_S^{(1)}\right) = \left(\mathbf{G}_S^{(1)} - E\left[\mathbf{G}_S^{(1)}\right]\right)^4 \quad (30)$$

where $E(\cdot)$ is the mathematical expectation. Meanwhile, the scaling matrix $\mathbf{H} \in \mathbb{R}^{X \times M}$ between $\mathbf{G}_T$ and $\mathbf{G}_{T_S}$ is computed as

$$\mathbf{H} = \mathbf{G}_{T_S}^{(1)} \times \mathrm{Pre}\left(\mathbf{G}_T^{(1)}\right) \quad (31)$$

where $\mathrm{Pre}(\cdot)$ calculates the pseudoinverse. Similarly, the aligned 3DDGG feature data $\widehat{\mathbf{G}_S^{(1)}} \in \mathbb{R}^{X \times (Y \times B_S)}$ and $\widehat{\mathbf{G}_{T_S}^{(1)}} \in \mathbb{R}^{X \times (P \times B_T)}$ using the fourth-order statistics can be acquired

$$\widehat{\mathbf{G}_S^{(1)}} = \left(\mathbf{K}_S^{-\frac{1}{2}} \mathbf{K}_{T_S}^{\frac{1}{2}}\right)^T \mathbf{G}_S^{(1)} \quad (32)$$

$$\widehat{\mathbf{G}_{T_S}^{(1)}} = \left(\mathbf{K}_{T_S}^{-\frac{1}{2}} \mathbf{K}_S^{\frac{1}{2}}\right)^T \mathbf{G}_{T_S}^{(1)}. \quad (33)$$

Then, the scaling matrix $\mathbf{H}$ is used to restore the size of $\widehat{\mathbf{G}_{T_S}^{(1)}}$, and the aligned 3DDGG feature in the target domain $\widehat{\mathbf{G}_T^{(1)}} \in \mathbb{R}^{M \times (P \times B_T)}$ is obtained by

$$\widehat{\mathbf{G}_T^{(1)}} = \mathrm{Pre}(\mathbf{H}) \times \widehat{\mathbf{G}_{T_S}^{(1)}}. \quad (34)$$

Finally, the aligned data of source and target domain, $\widehat{\mathbf{G}_S^{(1)}}$ and $\widehat{\mathbf{G}_T^{(1)}}$, are folded into 3-D domain, which are denoted as $\widehat{\mathcal{G}_S^{(1)}} \in \mathbb{R}^{X \times Y \times B_S}$ and $\widehat{\mathcal{G}_T^{(1)}} \in \mathbb{R}^{M \times P \times B_T}$. Likewise, the aligned data on the other two dimensions (the y-axis and the spectral dimension) can be attained in a similar way. $\widehat{\mathcal{G}_S^{(2)}} \in \mathbb{R}^{X \times Y \times B_S}$ and $\widehat{\mathcal{G}_T^{(2)}} \in \mathbb{R}^{M \times P \times B_T}$ and $\widehat{\mathcal{G}_S^{(3)}} \in \mathbb{R}^{X \times Y \times B_S}$ and $\widehat{\mathcal{G}_T^{(3)}} \in \mathbb{R}^{M \times P \times B_T}$, respectively, stand for the aligned data of source and target domain in the vertical and spectral directions.

Meanwhile, the second-order statistics used in traditional CORAL is also considered. It is easy to implement by just replacing the Cok function in (30) with the Cov function in (12). Correspondingly, the aligned data of source and

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target domain using the second-order statistics in all the three directions are, respectively, $\overline{\mathcal{G}_S^{(i)}} \in \mathbb{R}^{X\times Y\times B_S}$ and $\overline{\mathcal{G}_T^{(i)}} \in \mathbb{R}^{M\times P\times B_T}, i = 1, 2, 3$. On the other hand, the 3DSGG feature cubes in source and target domain, that is, $\mathbf{V}_S \in \mathbb{R}^{X\times Y\times B_S}$ and $\mathbf{V}_T \in \mathbb{R}^{M\times P\times B_T}$, can be managed in the same way. The resultant 12 feature modules (half is from the source domain, while the other half is from the target domain) are expressed as $\widehat{\mathcal{V}_S^{(i)}}$, $\overline{\mathcal{V}_S^{(i)}}$, $\widehat{\mathcal{V}_T^{(i)}}$, $\overline{\mathcal{V}_T^{(i)}}$, $i = 1, 2, 3$, and each one has the same size of the corresponding feature cube.

D. ELM-Based Multitask Fusion

So far, we have obtained a number of feature modules from the source and target domains. They can be presented as $\mathcal{P} = \{(\widehat{\mathcal{V}_S^{(i)}}, \widehat{\mathcal{V}_T^{(i)}}), (\overline{\mathcal{V}_S^{(i)}}, \overline{\mathcal{V}_T^{(i)}}), (\widehat{\mathcal{G}_S^{(i)}}, \widehat{\mathcal{G}_T^{(i)}}), (\overline{\mathcal{G}_S^{(i)}}, \overline{\mathcal{G}_T^{(i)}}), i = 1, 2, 3\}$. Due to the complementary properties of extracted features (demonstrated in Section IV), it is desirable to manipulate the dozen feature cubes in a multitask learning framework rather than directly concatenating them together.

In the domain adaptation scenario, most training samples are selected from the source domain $\mathbf{R}_S$, and limited samples are chosen from the target domain $\mathbf{R}_T$. Similar to the description in Section II-E, let $\mathbf{A} = \{(\mathbf{x}_i, \mathbf{y}_i)|\mathbf{x}_i \in \{\mathbf{R}_S, \mathbf{R}_T\}\}_{i=1}^N$ represent the training set with N samples, where $\mathbf{x}_i \in \mathbb{R}^B$, if $B_S < B_T$, $B = B_S$; otherwise, $B = B_T$ (the scaling method introduced in Section III-C is used to scale B_S or B_T to B), is the feature vector and $\mathbf{y}_i \in \mathbb{R}^C$ is the corresponding label vector (C is the class number). With respect to the achieved multiple feature modules, the corresponding training set is obtained by

$$\mathcal{A}_j = \left\{\left(\mathbf{x}_i^*, \mathbf{y}_i^*\right)\middle|\mathbf{x}_i^* \in \mathcal{P}_j, \mathrm{Pos}(x_i) = \mathrm{Pos}\left(\mathbf{x}_i^*\right)\right\} \quad i = 1, \ldots, N, j = 1, \ldots, 12 \tag{35}$$

where $\mathrm{Pos}(\cdot)$ is the spatial position of each sample. It is worth mentioning that a class center alignment operation [59], which moves each training sample in the source domain to the target domain by aligning the mean spectrum of every class, is performed to further decrease the discrepancy of the training samples in source domain. After applying the ELM learning algorithm [see (17)] on the aligned training set $\widetilde{\mathcal{A}}_j$ of all the 12 modules, the related parameters, including $\beta_{j,k}$, $\mathbf{W}_{j,k}$, and $\mathbf{b}_{j,k}$, $(j = 1, \ldots, 12, k = 1, \ldots, K)$ (recall that K is number of hidden nodes in the ELM model), can be estimated. Finally, the multitask fusion scheme is employed to accomplish the classification task

$$\mathcal{L}_\mathbf{z} = \underset{c=1,\ldots,C}{\arg\max} \sum_{j=1}^{12}\sum_{k=1}^{K} \beta_{j,k} f\left(\mathbf{W}_{j,k} \cdot \mathbf{z}^* + \mathbf{b}_{j,k}\right) \tag{36}$$

where $\mathbf{z}$ is any test sample and $\mathbf{z}^*$ stands for the corresponding feature in each module. The pseudocode of our GF-3DDA approach for HSI classification is provided in Algorithm 1.

IV. Experiment Setup and Ablation Analysis

First, three real-world hyperspectral data sets that each contains two scenes are introduced. Second, several issues in experiment design, including sample partition and evaluation metrics, are presented. Third, parameter analysis is provided. Finally, module ablation analysis is carried out to demonstrate the effectiveness of the proposed GF-3DDA approach.

Algorithm 1 GF-3DDA for HSI Classification

1: **INPUT**: Raw hyperspectral image $\mathbf{R}_S \in \mathbb{R}^{X\times Y\times B_{S'}}$ and $\mathbf{R}_T \in \mathbb{R}^{M\times P\times B_{T'}}$ in source domain and target domain, training set $\mathbf{A} = \{(\mathbf{x}_i, \mathbf{y}_i)|\mathbf{x}_i \in \{\mathbf{R}_S, \mathbf{R}_T\}\}_{i=1}^N$, test set $\mathbf{Z} \in \mathbb{R}^{B\times Q}$;
2: **OUTPUT**: Predicted labels $\mathcal{L} \in \mathbb{R}^Q$;
3: **BEGIN**
4: Using 3D Gabor to filter $\mathbf{R}_S$ and $\mathbf{R}_T$ separately to get the $\mathbf{R}_{GS} \in \mathbb{R}^{X\times Y\times B_S}$ and $\mathbf{R}_{GT} \in \mathbb{R}^{M\times P\times B_T}$
5: Using equation (23) to generate the 3DSGG feature data $\mathbf{V}_S$ and $\mathbf{V}_T$, and using equation (27) to get the 3DDGG feature $\mathbf{G}_S$ and $\mathbf{G}_T$;
6: Using the proposed 3DBCORAL-SF method to achieve a dozen domain-adapted feature cubes $\mathcal{P}$;
7: Using equation (35) to obtain the training set $\mathcal{A}_j$, $j = 1, \ldots, 12$ in each module, which is then aligned and utilized to train the ELM model;
8: For each test sample $\mathbf{z} \in \mathbb{R}^B$ in $\mathbf{Z}$, using equation (36) to predict its label.
9: **END**

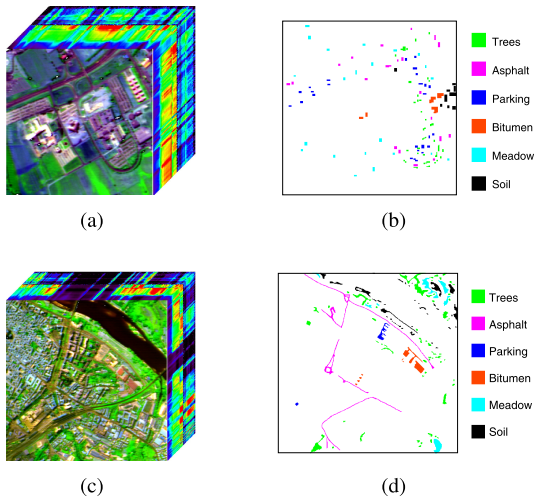


Fig. 3. Source and target Pavia HSI data and the ground-truth maps. Source data cube of (a) Pavia University and (b) ground-truth map. Target data cube of (c) Pavia center and (d) ground-truth map.

A. Experimental Data

For the first experiment, the two Pavia HSI data collected by the airborne sensor DAIS in Pavia, Italy, are employed. The original size of the Pavia university scene is $243 \times 243 \times 72$, which is considered as the source data. A cropped Pavia center image was selected as the target image with a size of $400 \times 400 \times 72$. The spatial resolution of the two data sets is 1.3 m/pixel. The source and target hyperspectral data cubes and ground-truth maps are shown in Fig. 3. Six land cover classes contained in both data sets are selected for classification, and the detailed information is listed in Table II.

The second source and target scenes of HSI were, respectively, obtained in Shanghai and Hangzhou, China, and both data sets were captured by Hyperion hyperspectral sensor EO-1. The source scene was taken from Shanghai, the largest city in the east of China, on April 1, 2002. Its size is

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TABLE II

NUMBER OF LABELED SAMPLES OF EACH LAND COVER CLASS IN THE SOURCE AND TARGET PAVIA HSI DATA

Class	Land cover type	No. of labeled samples in Pavia university (source)	No. of labeled samples in Pavia center (target)
$C1$	Trees	266	2424
$C2$	Asphalt	266	1704
$C3$	Parking lot	265	287
$C4$	Bitumen	206	685
$C5$	Meadow	273	1251
$C6$	Soil	213	1475
	Total	1489	7826

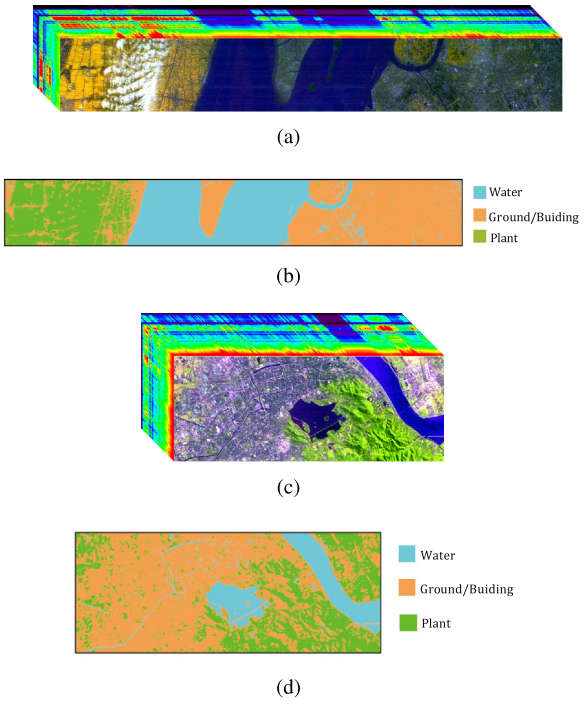


Fig. 4. Source and target Shanghai–Hangzhou HSI data and the ground-truth maps. Source data cube of (a) Shanghai and (b) ground-truth map. Target data cube of (c) Hangzhou and (d) ground-truth map.

$1600 \times 230 \times 198$, and the land surface mainly includes rivers, buildings, plants, and roads. The target data were collected on November 2, 2002 over Hangzhou, the capital city of Zhejiang province in China, with similar land covers as Shanghai. The size of the Hangzhou scene (taken as the target domain) is $590 \times 230 \times 198$. Compared with the aforementioned Pavia HSI, the spatial resolution of Shanghai–Hangzhou data sets is much lower, which is only 30 m/pixel. Similarly, the data cubes and ground-truth maps are shown in Fig. 4, and the number of labeled samples of each land cover class is given in Table III.

The third HSI data set was captured by the AVIRIS sensor at Northwest Indiana on June 12, 1992. The size of the entire Indiana image is $614 \times 2166 \times 220$, covering over 25×6 mi on the surface. Two unconnected areas were cut out from the Indiana HSI, which are, respectively, named Indiana-P1 and Indiana-P2. Both data have the same size, i.e., $300 \times 400 \times 220$. Since the Indiana image is collected almost 30 years

TABLE III

NUMBER OF LABELED SAMPLES OF EACH LAND COVER CLASS IN THE SOURCE AND TARGET SHANGHAI–HANGZHOU HSI DATA

Class	Land cover type	No. of labeled samples in Shanghai (source)	No. of labeled samples in Hangzhou (target)
$C1$	Water	123123	18043
$C2$	Land/Building	161689	77450
$C3$	Plant	83188	40207
	Total	368000	135700

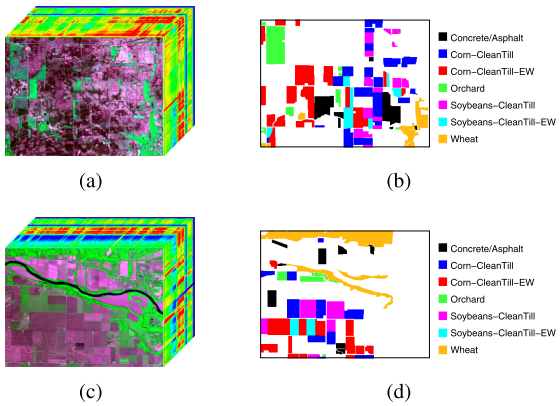


Fig. 5. Source and target Indiana HSI data and the ground-truth maps. Source data cube of (a) Indiana-P1 and (b) ground-truth map. Target data cube of (c) Indiana-P2 and (d) ground-truth map.

TABLE IV

NUMBER OF LABELED SAMPLES OF EACH LAND COVER CLASS IN THE SOURCE AND TARGET INDIANA HSI DATA

Class	Land cover type	No. of labeled samples in Indiana-P1 (source)	No. of labeled samples in Indiana-P2 (target)
$C1$	Concrete/Asphalt	4867	2942
$C2$	Corn-CleanTill	9822	6029
$C3$	Corn-CleanTill-EW	11414	7999
$C4$	Orchard	5106	1562
$C5$	Soybeans-CleanTill	4731	4792
$C6$	Soybeans-CleanTill-EW	2996	1638
$C7$	Wheat	3223	10739
	Total	42159	35701

ago, the spatial resolution is only 30 m/pixel, which is the same as Shanghai–Hangzhou data sets. The data cubes and related ground-truth maps are shown in Fig. 5. Seven land cover classes are provided for classification, and the number of labeled samples is shown in Table IV.

The fourth HSI data set comes from the Houston University that is captured in different years, i.e., 2013 and 2018, which provides a very good scenario for domain adaptation. Specifically, the Houston 2013 hyperspectral data set was collected on June 23, 2012 with the Compact Airborne Spectrographic Imager (CASI) at the University of Houston and nearby areas. The data size is $349 \times 1905 \times 144$, that is, the space size is 349×1905, and it has 144 spectral bands with the 2.5-m spatial solution. Ten categories are selected for training and testing.

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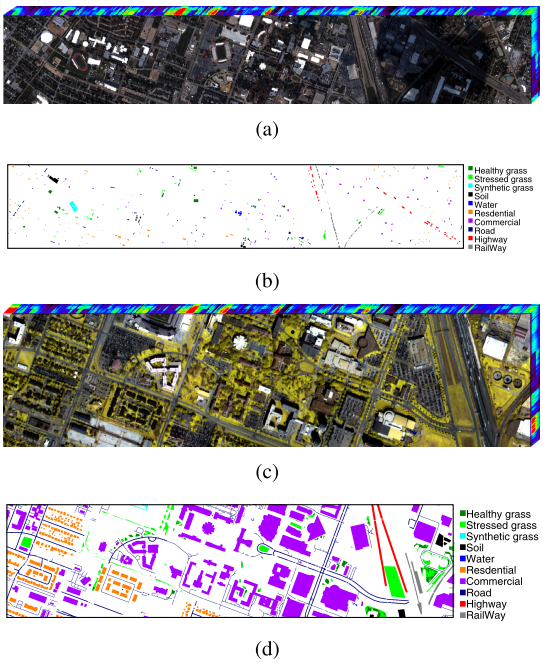


Fig. 6. (a) HSI data and (b) ground-truth map of Houston 2013 HSI. (c) HSI data and (d) ground-truth map of Houston 2018 HSI.

TABLE V

NUMBER OF LABELED SAMPLES OF EACH LAND COVER CLASS IN THE SOURCE AND TARGET HOUSTON HSI DATA

Class	Land cover type	No. of labeled samples in Houston2018 (Source)	No. of labeled samples in Houston2013 (Target)
$C1$	Health grass	9799	1251
$C2$	Stressed grass	32502	1254
$C3$	Synthetic grass	684	697
$C4$	Soil	4516	1242
$C5$	Water	266	325
$C6$	Residential	39762	1268
$C7$	Commercial	223684	1244
$C8$	Road	45810	1252
$C9$	Highway	9849	1227
$C10$	Railway	6937	1235
	Total	373809	10995

The hyperspectral data map and ground truth of Houston University 2013 are shown in Fig. 6(a) and (b), respectively. The label categories are shown in Table V. Alternatively, Houston 2018 is a relatively new data, which was collected by the University of Houston on February 16, 2017 with a hyperspectral imager CASI 1500. The entire data set of Houston 2018 was distributed by the Image Analysis and Data Fusion Technical Committee of the Geographical Sciences and Remote Sensing Society (GRSS) and the University of Houston in the 2018 Data Fusion Competition. The data size is $601 \times 2384 \times 48$, and its sampling distance from the ground is 1 m. Similarly, ten categories are chosen. The hyperspectral data map and the ground truth are shown in Fig. 6(c) and (d), respectively, and the label classes are shown in Table V. Since there are more labeled samples in the Houston 2018 data set, it is taken as the source domain data and the target domain data is Houston 2013.

TABLE VI

NUMBER OF LABELED SAMPLES PER CLASS FOR TRAINING IN SOURCE AND TARGET DOMAINS

Source domain	Pavia University	180
	Shanghai	540
	Indiana-P1	200
	Houston 2018	200
Target domain	{5,10,15,20,25}	

B. Experiment Design

In the aspect of experiment design, the significance of practical application is fully considered. Generally, there are only a few labeled samples in the target scene in practice. Fortunately, in the domain adaptation scenario, sufficient labeled samples existed in the source domain can be imported and have a positive impact on the classification performance. Hence, the training set consists of a large number of labeled samples in the source domain and few labeled samples in the target domain, while the rest labeled samples in the target domain are regarded as the test set for classification. Specifically, a fixed number of each class is randomly selected from the source and target domains to constitute the training set. Since the number of labeled samples in four source HSI data is totally different (as listed in Tables II–V), the chosen training samples of each class also have various numbers, which are, respectively, 180, 540, 200, and 200 for the Pavia University, Shanghai, Indiana-P1, and Houston 2018 data sets. With respect to the target HSI data, the number of labeled samples per class for training varies from 5 to 25 with a step of 5. The detailed description of sample partition is shown in Table VI.

Alternatively, two popular evaluation metrics, both overall accuracy (OA) and Kappa coefficient, are utilized in the experiments. OA is calculated according to the corrected classified samples, whereas Kappa is computed based on the confusion matrix. Both metrics, respectively, reflect the global and statistical performance of classification algorithm, and higher values correspond to better results. Since the small sample set problem is mainly concerned in this article, each experiment is repeated ten times, and both the mean and standard deviation of the two measures are reported.

C. Parameter Analysis

Practically, in the proposed GF-3DDA algorithm, partial parameters can be preset, such as the 3DSGG filters defined in Section III-A. Alternatively, the proposed 3DBCORAL-SF method is nonparametric, while the parameters of ELM classifier are optimized according to the original paper [44]. Concerning 3DDGG in Section III-B, it can be observed that parameter σ (standard deviation of Gaussian function) plays an essential role in the proposed method, which should be specially analyzed.

Fig. 7 shows the OA of 3DDGG method with different $\sigma = 2^m$ on the four HSIs, and m varies from 1 to 10 with a

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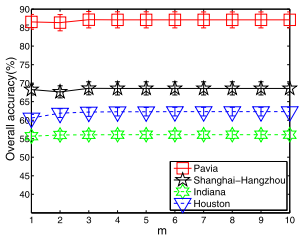


Fig. 7. OA of 3DDGG method with different $\sigma = 2^m$ when the number of labeled samples per class in target domain is 10. The horizontal axis is the value of m.

step of 1. Due to the low spatial resolution and complicated distribution of surface materials in the Indiana HSI data set, the OA on the Indiana data is much lower than the other two data sets. Meanwhile, it can be observed from Fig. 7 that the OA measure is almost stable with various m (the Gaussian kernel is normalized after using each parameter m to produce the Gaussian kernel, which prevents the Gaussian kernel from being infinitely small and make our Gaussian kernel uniform), implying the robustness of introduced 3DDGG feature extraction procedure. In all experiments, parameter m is simply set as 3, and σ equals 8 correspondingly. In summary, since the parameters contained in our GF-3DDA approach can be either kept unchanged for various data sets or learned in the training process, the generalization ability of the method can be ensured.

D. Module Ablation Analysis

A number of ablation methods are designed in comparison with the proposed GF-3DDA algorithm, in which several methods are presented as follows.

1) RAW is designed as applying the ELM model on the original HSI data sets. The training set contains samples from both source and target domains, which is used to train the ELM model, and then, the optimized ELM model is applied on the test set in the target domain. Clearly, the RAW method is carried out without feature extraction and domain adaptation, which is considered as the benchmark method.
2) 3DSGG is the method that both the training and test sets are chosen from the achieved 3DSGG features, while other steps are the same as the RAW method.
3) 3DSGG-3DDA employs the proposed 3DBCORAL-SF domain adaptation process on the 3DSGG feature. As presented in Section III-C, all three orientations (two spatial and spectral) and two statistical measures (second and fourth orders) are investigated in domain adaptation, eventually obtaining six modules in total $\{(\widehat{\mathcal{V}_S^{(i)}}, \widehat{\mathcal{V}_T^{(i)}}), (\overline{\mathcal{V}_S^{(i)}}, \overline{\mathcal{V}_T^{(i)}}), i = 1, 2, 3\}$. Similarly, the probabilistic fusion manner is used to carry out sample classification.

With respect to the other designed 3DDGG feature, two 3DSGG-based compared methods, including 3DSGG and 3DSGG-3DDA, can be updated to two 3DDGG-based ones by simply replacing the 3DSGG feature with the 3DDGG feature, which are, respectively, named 3DDGG and 3DDGG-3DDA.

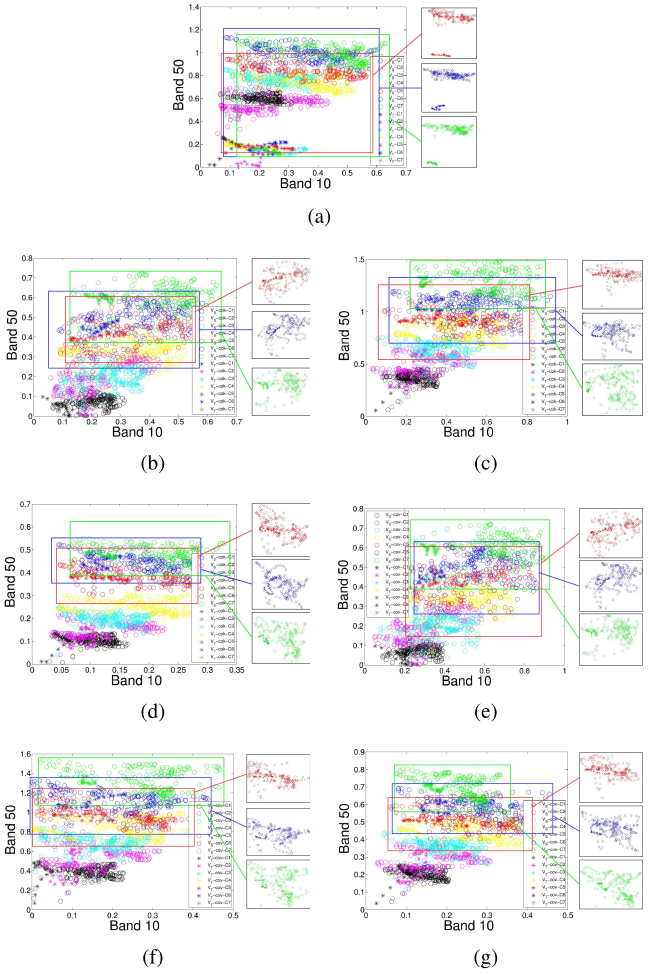


Fig. 8. Scatter plot of training samples selected from Indiana-P1 (source) and Indiana-P2 (target) HSI data sets in different 3DSGG-based modules. Here, bands 10 and 50 are chosen as the two axes. (a) Training samples in 3DSGG feature $\{\mathbf{V}_S, \mathbf{V}_T\}$. (b) $\widetilde{\mathcal{A}}_1$. (c) $\widetilde{\mathcal{A}}_2$. (d) $\widetilde{\mathcal{A}}_3$. (e) $\widetilde{\mathcal{A}}_4$. (f) $\widetilde{\mathcal{A}}_5$. (g) $\widetilde{\mathcal{A}}_6$. Here, circles "○" with various colors indicate the samples from the source domain, whereas stars "*" with various colors indicate the samples from the target domain.

Meanwhile, it can be easily found that three methods (RAW, 3DSGG, and 3DDGG) are nondomain-adaptation techniques, while the rest two methods (3DSGG-3DDA and 3DDGG-3DDA) include domain adaptation operation. Finally, the multitask fusion scheme is utilized to combine 3DSGG, 3DDGG, 3DSGG-3DDA, and 3DDGG-3DDA together to achieve the classification results of the proposed GF-3DDA.

Before presenting the classification results of compared ablation methods, the distinguishing characteristics of each part in GF-3DDA, including 3DSGG and 3DDGG feature extraction and 3DBCORAL-SF domain adaptation, are visually examined. Taking the Indiana hyperspectral data sets as an example (which contains seven classes), Figs. 8 and 9 show the scatter plot of training samples selected from Indiana-P1 (source, 200 samples per class) and Indiana-P2 (target, 25 samples per class) HSI data sets in different 3DSGG- and 3DDGG-based modules. It can be clearly seen from Figs. 8(a) and 9(a) that the data distribution between the same category in the source and target domains is very different, such as the three classes that are enlarged in red,

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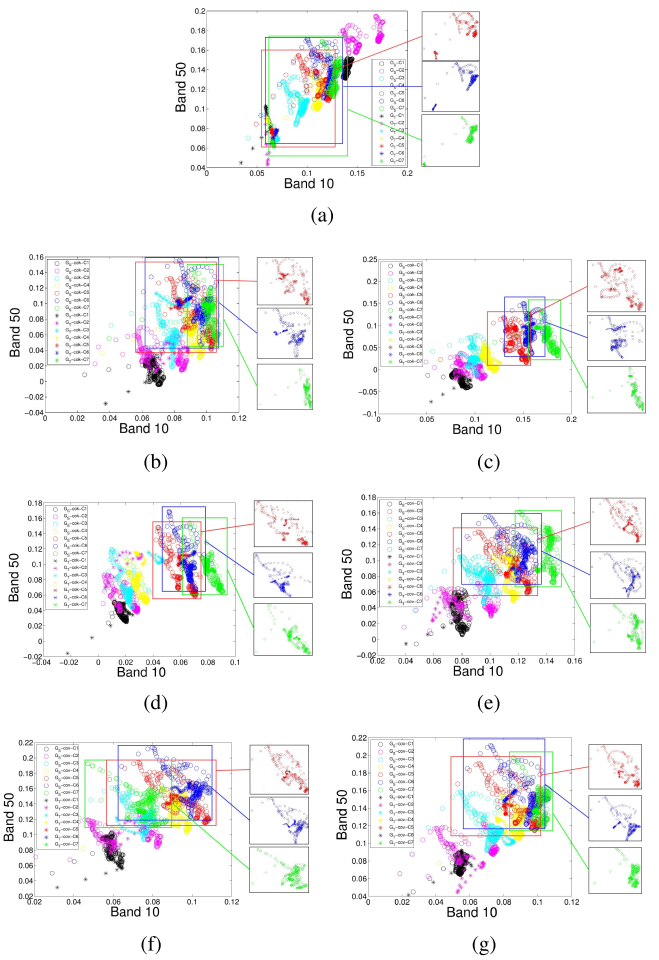


Fig. 9. Scatter plot of training samples selected from Indiana-P1 (source) and Indiana-P2 (target) HSI data sets in different 3DDGG-based modules. Here, bands 10 and 50 are chosen as the two axes. (a) Training samples in 3DDGG feature $\{\mathbf{G}_S, \mathbf{G}_T\}$. (b) $\widetilde{\mathcal{A}}_7$. (c) $\widetilde{\mathcal{A}}_8$. (d) $\widetilde{\mathcal{A}}_9$. (e) $\widetilde{\mathcal{A}}_{10}$. (f) $\widetilde{\mathcal{A}}_{11}$. (g) $\widetilde{\mathcal{A}}_{12}$. Here, circles "○" with various colors indicate the samples from the source domain, whereas stars "*" with various colors indicate the samples from the target domain.

blue, and green colors. Meanwhile, the simple alignment operator can make the samples from the same class in the source and target domains be more consistent, as shown in Figs. 8(b)–(g) and 9(b)–(g). Besides, the data distribution of training samples in Figs. 8(b)–(g) and 9(b)–(g) is different from each other, indicating the complementary properties of the large amount of extracted feature modules.

The results are shown in Table VII. It can be easily seen from Table VII that the classification accuracies of RAW method in all circumstances are the lowest, implying the importance of feature extraction. Meanwhile, the performance of three domain adaptation-based methods, including 3DSGG-3DDA, 3DDGG-3DDA, and proposed GF-3DDA, is generally better than that of nondomain-adaptation ones (RAW, 3DSGG, and 3DDGG), indicating the necessity of domain adaptation procedure. In particular, our GF-3DDA approach shows the best results for all the four data sets, validating rationality of simultaneously employing 3DSG and 3DDtG feature cubes.

TABLE VII

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE FOUR HYPERSPECTRAL DATA SETS WITH 180 (PAVIA UNIVERSITY), 540 (SHANGHAI), 200 (INDIANA-P1 AND HOUSTON 2018) LABELED SAMPLES PER CLASS OF SOURCE DOMAIN, AND TEN LABELED SAMPLES PER CLASS OF TARGET DOMAIN AS THE TRAINING SET

Data set	Algorithm	OA (%)	AA (%)	Kappa
Pavia	RAW	65.14	59.57	0.52
	3DSGG	84.79	84.19	0.81
	3DSGG-3DDA	88.60	88.58	0.85
	3DDGG	87.14	87.61	0.84
	3DDGG-3DDA	90.72	93.10	0.89
	GF-3DDA	**93.31**	**93.94**	**0.92**
Shanghai-Hangzhou	RAW	59.61	48.86	0.17
	3DSGG	76.45	80.93	0.61
	3DSGG-3DDA	78.90	80.65	0.65
	3DDGG	68.60	62.56	0.35
	3DDGG-3DDA	72.08	76.06	0.54
	GF-3DDA	**84.35**	**86.08**	**0.73**
Indiana	RAW	43.72	36.37	0.31
	3DSGG	56.42	50.62	0.46
	3DSGG-3DDA	62.27	63.65	0.54
	3DDGG	56.55	56.03	0.47
	3DDGG-3DDA	68.00	72.12	0.61
	GF-3DDA	**69.32**	**72.51**	**0.63**
Houston	RAW	56.85	60.78	0.52
	3DSGG	68.20	70.67	0.64
	3DSGG-3DDA	75.77	77.72	0.73
	3DDGG	63.12	65.95	0.58
	3DDGG-3DDA	73.16	75.22	0.70
	GF-3DDA	**80.06**	**78.96**	**0.77**

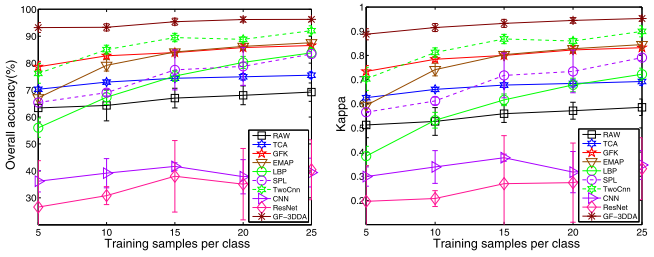


Fig. 10. Pavia HSI. (a) OA and (b) Kappa as functions of the number of training samples per class in target domain.

The computation time was recorded by a workstation with 24-core AMD processor of 2.20 GHz and 128-GB RAM. The running time of different methods, including the RAW method, two gradient feature extraction methods (3DSGG and 3DDGG), two domain adaptation methods (3DSGG-3DDA and 3DDGG-3DDA), and the proposed GF-3DDA method, is shown in Table VIII. It can be seen from the table that the computational cost of the proposed GF-3DDA method is applicable for HSI with large coverage.

V. EXPERIMENT RESULTS

In this section, we aim to demonstrate the superiority of the proposed GF-3DDA method. Eight recent and relevant state-of-the-art methods are taken into consideration, including two feature extraction methods (EMAPs [45] and LBP [46]), four domain adaptation methods (TCA [47], GFK [48], SPL [49], and TwoCnn [50]), and two deep learning classification

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TABLE VIII

RUNNING TIME (IN SECONDS) OF THE SIX COMPARED TECHNIQUES FOR THE FOUR HSIS WHEN TEN SAMPLES PER CLASS IN TARGET DOMAIN ARE USED FOR TRAINING

	RAW	3DSGG	3DSGG-3DDA	3DDGG	3DDGG-3DDA	GF-3DDA
Pavia	18.36	67.20	141.12	67.67	91.54	235.51
Shanghai-Hangzhou	123.43	132.11	226.51	122.61	170.34	446.34
Indiana	97.39	123.88	201.38	150.70	206.91	412.83
Houston	238.80	234.38	264.41	249.87	279.90	571.04

TABLE IX

PAVIA HSI: CLASSIFICATION PERFORMANCE OF EIGHT METHODS ON THE TEST SET WITH 180 LABELED SAMPLES PER CLASS OF SOURCE DOMAIN AND TEN LABELED SAMPLES PER CLASS OF TARGET DOMAIN AS THE TRAINING SET

	RAW		TCA		GFK		EMAP		LBP		SPL		TwoCnn		CNN		ResNet		GF-3DDA	
Class	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std
C1	88.33	8.07	90.18	2.12	**97.17**	0.54	80.01	2.31	66.98	6.17	95.46	0.42	87.77	5.09	50.57	7.49	26.93	39.28	92.96	2.54
C2	66.00	13.19	85.55	1.51	78.24	2.99	74.29	4.97	34.10	5.82	90.63	0.93	74.50	8.35	24.89	6.61	57.77	36.71	**96.97**	1.94
C3	59.39	17.09	84.40	5.04	82.02	3.26	**100.00**	0.00	97.47	5.05	89.65	0.78	88.45	4.86	72.26	8.19	9.03	21.49	99.77	0.51
C4	76.61	15.25	**93.78**	1.73	71.67	9.59	79.73	4.74	85.93	3.85	90.91	0.64	83.36	1.59	30.30	14.40	24.42	38.14	90.56	3.58
C5	33.10	25.50	47.74	1.66	88.46	2.77	93.13	4.37	87.28	5.71	43.14	3.34	94.20	2.80	20.15	5.67	38.96	40.58	**95.72**	2.89
C6	33.97	13.25	39.85	2.72	64.42	3.26	67.77	10.71	43.99	3.56	8.71	0.98	85.03	3.23	50.74	8.48	6.37	13.37	**87.66**	2.47
OA	65.14	5.75	73.00	0.60	82.72	0.80	79.24	1.94	67.45	2.07	69.08	0.89	85.02	1.15	39.15	5.80	30.85	3.90	**93.31**	1.29
Kappa	0.52	7.22	0.66	0.79	0.78	1.11	0.74	2.66	0.53	2.61	0.61	1.57	0.81	1.93	0.34	6.77	0.21	3.27	**0.92**	1.66

methods (CNN [51] and ResNet [52]). The performance of the proposed GF-3DDA approach and a number of alternatives on four HSI data sets is sequentially evaluated.

A. Pavia Data

Fig. 10 shows the OA and Kappa measure obtained by ten compared methods on Pavia HSI. As listed in Table VI, the number of labeled samples per class in the source scene (i.e., Pavia University) is 180, while labeled samples per class of target scene (i.e., Pavia Center) varies from 5 to 25. It can be easily seen from Fig. 10 that the classification accuracies of CNN and ResNet method in all circumstances are relatively low, implying the inefficiency of deep learning-based methods for small sample set problem, and even there are many labeled samples in the source domain. Meanwhile, the performance of four domain adaptation-based methods (TCA, GFK, SPL, and TwoCnn) and two feature extraction methods (EMAP and LBP) is better than that of RAW method, indicating the necessity of domain adaptation and feature extraction procedure. Furthermore, our GF-3DDA approach provides the best results all the way, demonstrating the effectiveness of the proposed method.

For the case when only ten labeled samples of each class in the target scene (Pavia Center) are chosen to constitute the training set, Table IX shows the classification accuracies of every class of each compared method, and OA and Kappa coefficient are also given in the meantime. In order to facilitate visual comparison, the classification maps of ten methods on the testing set in a single experiment are visualized in Fig. 11. Actually, the conclusion made in the last paragraph can also be drawn from Table IX and Fig. 11, and the superiority of our GF-3DDA approach is revealed.

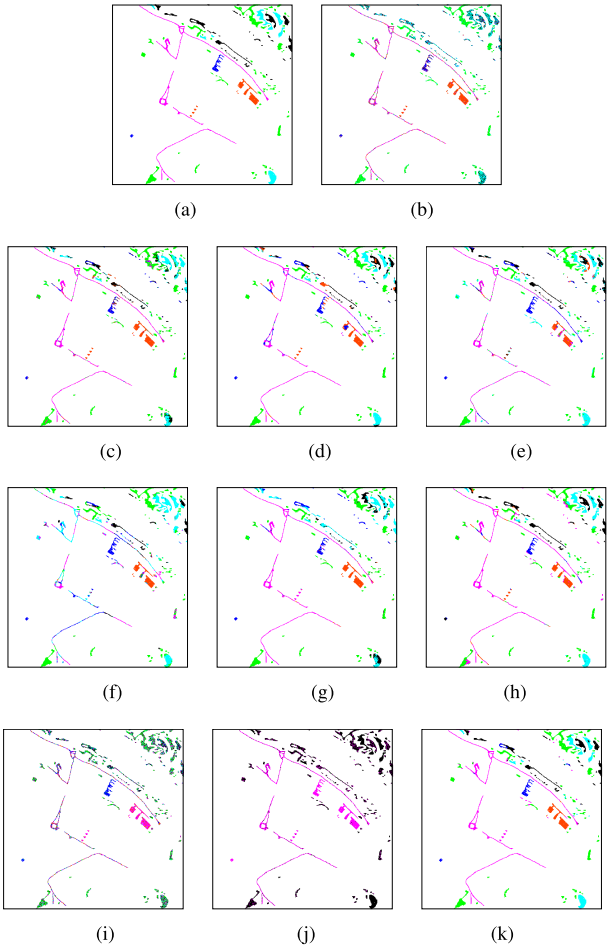


Fig. 11. Pavia Center HSI (ten labeled samples per class of the target domain for training): (a) testing map, (b) RAW (64.60%), (c) TCA (72.34%), (d) GFK (83.23%), (e) EMAP (78.14%), (f) LBP (67.49%), (g) SPL (70.22%), (h) TwoCnn (85.90%), (i) CNN (37.70%), (j) ResNet (34.64%), and (k) GF-3DDA (93.39%).

B. Shanghai–Hangzhou Data

Second, the Shanghai–Hangzhou HSI data are conducted for validation. Fig. 12 shows the OA and Kappa values of the ten compared methods with different training sets. Since the labeled samples in the source scene (Shanghai data) are huge, here the number of chosen labeled samples per class for training is 540. Moreover, the number of training samples

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TABLE X

SHANGHAI–HANGZHOU HSI: CLASSIFICATION PERFORMANCE OF EIGHT METHODS ON THE TEST SET WITH 540 LABELED SAMPLES PER CLASS OF SOURCE DOMAIN AND TEN LABELED SAMPLES PER CLASS OF TARGET DOMAIN AS THE TRAINING SET

	RAW		TCA		GFK		EMAP		LBP		SPL		TwoCnn		CNN		ResNet		GF-3DDA	
Class	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std
C1	54.81	30.02	98.02	1.84	97.56	0.89	**99.93**	0.15	66.16	1.42	92.49	0.01	99.65	0.21	78.41	8.25	69.25	6.38	94.12	2.44
C2	51.58	20.69	55.08	29.73	57.25	14.47	74.68	7.10	78.18	1.91	44.12	7.80	82.58	11.02	66.51	10.05	53.39	8.35	**84.54**	4.57
C3	40.15	0.42	**88.89**	8.01	81.81	4.47	71.09	0.61	37.44	2.70	58.21	0.54	15.03	2.94	77.53	10.59	66.04	8.94	79.58	7.69
OA	59.61	9.78	70.80	14.50	73.50	1.96	76.97	3.88	61.59	0.42	54.73	4.60	64.84	5.45	71.35	7.17	59.25	5.60	**84.35**	1.87
Kappa	0.17	10.71	0.57	19.47	0.58	3.41	0.63	5.42	0.24	0.55	0.30	8.13	0.36	8.08	0.66	7.32	0.52	4.35	**0.73**	3.28

TABLE XI

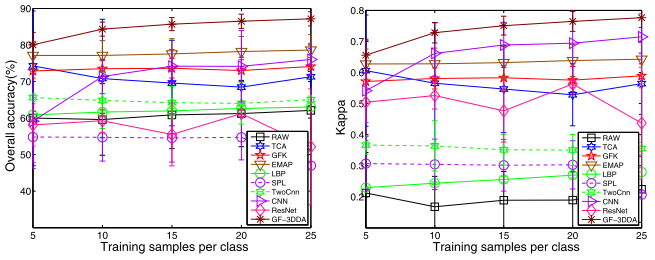
INDIANA HSI: CLASSIFICATION PERFORMANCE OF EIGHT METHODS ON THE TEST SET WITH 200 LABELED SAMPLES PER CLASS OF SOURCE DOMAIN AND TEN LABELED SAMPLES PER CLASS OF TARGET DOMAIN AS THE TRAINING SET

	RAW		TCA		GFK		EMAP		LBP		SPL		TwoCnn		CNN		ResNet		GF-3DDA	
Class	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std
C1	32.14	20.11	52.97	7.80	40.10	5.12	56.10	7.87	32.31	5.15	15.31	0.44	40.35	4.63	43.30	17.20	38.23	42.50	**71.77**	8.79
C2	12.98	4.28	14.28	4.00	31.82	6.95	32.85	2.01	**57.35**	5.45	26.83	1.97	35.21	13.00	17.47	9.64	39.57	38.24	45.59	16.21
C3	37.10	23.58	20.32	1.91	22.11	2.24	37.05	7.36	**50.21**	5.19	23.42	0.10	18.88	2.66	10.34	7.68	5.93	7.70	45.24	14.79
C4	70.25	8.29	79.50	5.96	86.57	3.20	67.50	6.02	74.10	7.90	96.28	0.15	85.87	7.10	31.50	9.05	25.57	37.35	**96.80**	1.83
C5	13.95	1.83	34.44	5.48	31.41	1.90	54.07	11.05	47.06	7.39	25.71	25.77	40.86	12.98	37.59	15.67	17.30	2.65	**63.94**	13.72
C6	15.15	2.70	30.87	3.24	28.72	2.29	78.34	7.16	70.77	5.71	24.98	13.71	49.73	7.15	24.49	5.50	13.05	18.51	**88.88**	6.63
C7	73.01	9.80	**96.77**	0.21	91.58	1.87	72.09	4.34	85.19	2.39	90.29	0.57	81.45	4.63	67.72	10.63	20.31	21.51	93.35	1.68
OA	43.72	5.63	49.96	0.74	50.51	1.57	53.96	1.68	61.73	1.51	41.78	2.54	49.52	2.39	36.76	4.42	18.62	8.20	**69.32**	4.00
Kappa	0.31	5.12	0.39	0.98	0.40	2.17	0.45	2.09	0.51	2.37	0.31	3.55	0.39	3.76	0.33	4.26	0.11	9.09	**0.63**	4.78

TABLE XII

HOUSTON HSI: CLASSIFICATION PERFORMANCE OF EIGHT METHODS ON THE TEST SET WITH 200 LABELED SAMPLES PER CLASS OF SOURCE DOMAIN AND TEN LABELED SAMPLES PER CLASS OF TARGET DOMAIN AS THE TRAINING SET

	RAW		TCA		GFK		EMAP		LBP		SPL		TwoCnn		CNN		ResNet		GF-3DDA	
Class	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std	Mean	std
C1	85.37	2.32	68.30	9.48	78.66	13.55	84.93	2.66	58.20	2.56	73.06	13.11	73.17	3.03	41.71	11.97	6.80	9.73	**96.37**	4.46
C2	27.01	27.63	65.39	8.28	79.21	6.49	86.06	7.90	52.33	9.48	54.31	8.71	67.55	6.94	36.21	8.92	38.89	35.14	**87.46**	0.68
C3	98.28	1.57	90.04	2.37	99.65	0.17	99.64	0.17	99.51	0.58	**100.00**	0.00	90.93	0.76	66.30	17.89	30.54	35.02	99.85	0.12
C4	93.54	2.41	73.91	8.02	99.61	0.42	91.07	8.83	65.62	19.83	99.05	0.60	72.29	14.91	25.69	10.87	30.57	30.96	**100.00**	0.00
C5	84.95	0.65	85.24	4.95	76.25	0.86	83.17	5.93	91.93	5.39	72.06	4.61	81.16	4.07	70.79	12.31	6.93	16.64	**97.46**	3.94
C6	24.79	28.08	61.76	9.20	60.67	11.60	83.72	4.08	72.05	9.13	66.00	12.18	70.61	2.27	39.49	10.06	21.48	22.64	**85.69**	5.93
C7	42.51	7.74	58.23	4.36	55.56	9.75	59.66	2.83	39.72	3.61	31.50	11.19	39.47	8.80	33.58	8.54	24.96	34.27	**67.34**	6.92
C8	**84.48**	4.74	72.19	4.00	30.47	19.59	84.01	1.36	41.06	4.81	57.65	1.23	58.11	7.73	26.52	7.90	4.97	7.80	74.40	4.02
C9	13.47	8.68	50.48	6.05	52.33	3.08	60.48	9.93	**75.43**	9.50	4.55	3.86	69.65	10.10	34.47	18.56	19.21	2.33	34.84	5.10
C10	58.19	11.48	44.26	6.58	32.77	8.71	66.27	13.48	**87.54**	10.61	43.43	2.81	54.39	4.48	27.33	11.39	14.07	28.49	78.12	5.52
OA	56.85	8.19	64.32	1.87	64.06	2.76	78.07	1.83	64.67	1.59	57.26	0.80	65.45	1.51	35.99	4.08	18.47	5.85	**80.06**	1.55
Kappa	0.52	10.24	0.60	2.30	0.59	3.42	0.75	2.30	0.60	2.00	0.52	1.26	0.61	2.34	0.37	3.31	0.13	7.58	**0.77**	1.95



(a) (b)

Fig. 12. Shanghai–Hangzhou HSI. (a) OA and (b) Kappa as functions of the number of training samples per class in target domain.

per class in the target scene ranges from 5 to 25. As can be seen from Fig. 12, compared with the other nine compared methods, the classification result of GF-3DDA has obvious superiority. Likewise, the detailed comparison of ten methods is listed in Table X when ten labeled samples per class in the target scene (Hangzhou data) are chosen for training, while the classification maps of compared methods are shown in Fig. 13, and the same conclusion as the above experiment can be drawn.

C. Indiana Data

Third, the classification results of two measures (OA and Kappa) of RAW, TCA, GFK, EMAP, LBP, SPL, TwoCnn, CNN, ResNet, and proposed GF-3DDA methods on Indiana HSI scenes are shown in Fig. 14. Due to the low spatial resolution and complicated distribution of materials, the performance of all compared methods on the Indiana data is much lower than that of the above two HSI data sets. Here, the training samples of each class from the source domain (Indiana-P1) are 200 and the range of training samples per class from the target domain (Indiana-P2) is the same as the above experiments.

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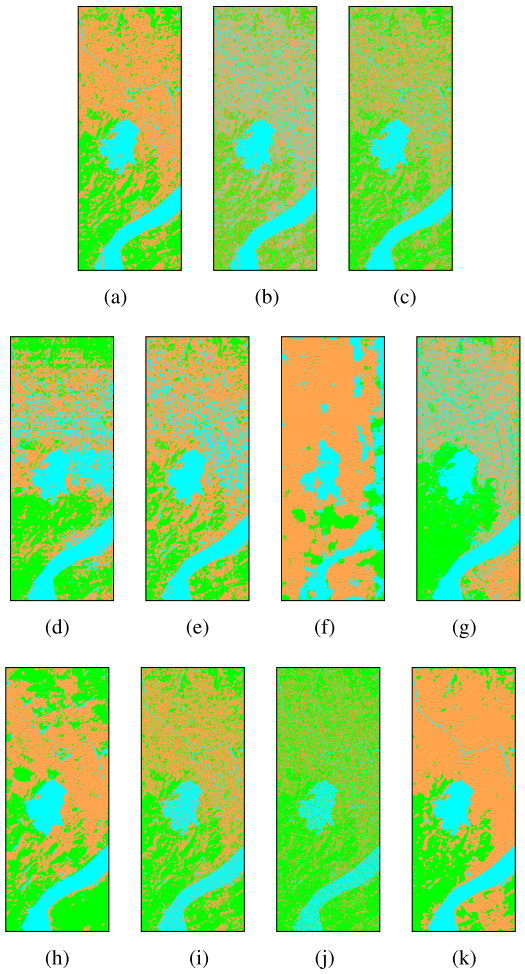


Fig. 13. Hangzhou HSI (ten labeled samples per class of the target domain for training): (a) testing map, (b) RAW (57.87%), (c) TCA (78.05%), (d) GFK (72.23%), (e) EMAP (73.69%), (f) LBP (58.81%), (g) SPL (58.88%), (h) TwoCnn (66.53%), (i) CNN (79.13%), (j) ResNet (67.09%), and (k) GF-3DDA (84.52%).

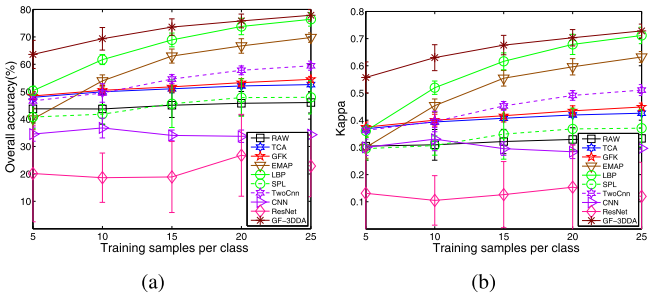


Fig. 14. Indiana HSI. (a) OA and (b) Kappa as functions of the number of training samples per class in target domain.

At the same time, the detailed classification accuracy results of the ten methods are shown in Table XI, in which the number of training samples per class in the target scene is 10. From the data in Table XI, we can see that of the ten methods, our GF-3DDA outputs the highest OA and Kappa values. Similarly, the classification maps of different methods on the testing set are shown in Fig. 15.

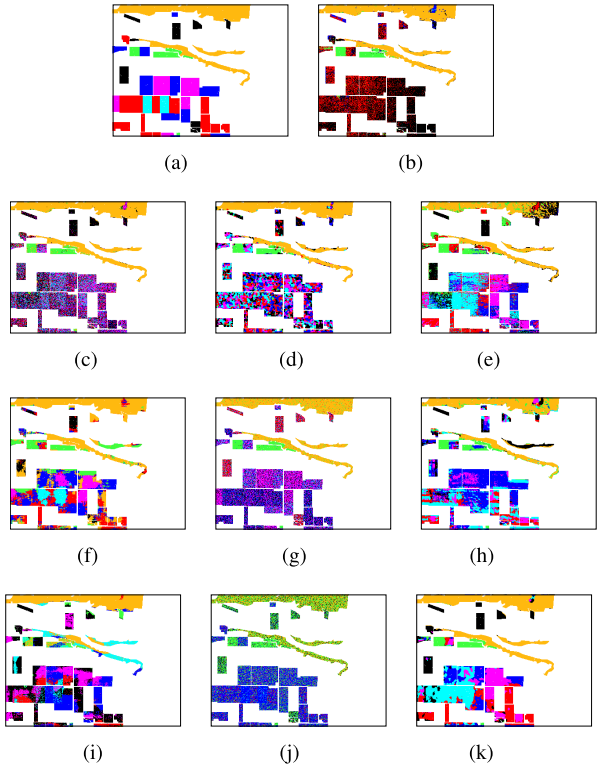


Fig. 15. Indiana-P2 HSI (ten labeled samples per class of the target domain for training): (a) testing map, (b) RAW (41.25%), (c) TCA (50.16%), (d) GFK (50.03%), (e) EMAP (53.66%), (f) LBP (62.52%), (g) SPL (45.92%), (h) TwoCnn (49.00%), (i) CNN (41.47%), (j) ResNet (34.24%), and (k) GF-3DDA (70.42%).

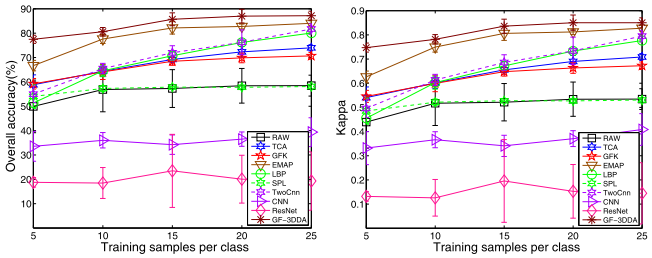


Fig. 16. Houston HSI. (a) OA and (b) Kappa as functions of the number of training samples per class in target domain.

D. Houston Data

At last, for the Houston HSI data sets, Houston 2018 and 2013 are, respectively, used as the source and target data. Because the spectral bands of the two data sets are different, and the acquisition time and other environmental factors are totally different, the Houston data are a very good candidate for domain adaptation scenario. Likewise, the detailed comparison results of each method are shown in Fig. 16 and Table XII. At the same time, the classification maps of various methods are shown in Fig. 17. It can be concluded from these quantitative and visual results that our GF-3DDA approach provides an excellent alternative for HSI classification.

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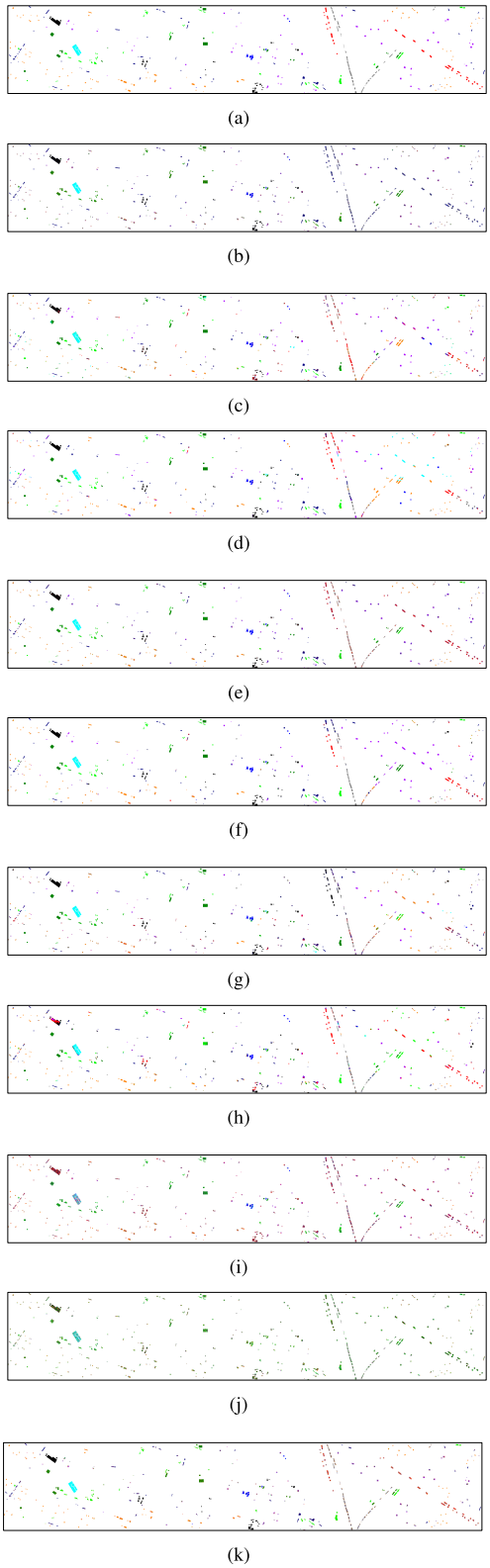


Fig. 17. Houston 2013 HSI (ten labeled samples per class of the target domain for training): (a) testing map, (b) RAW (51.59%), (c) TCA (60.79%), (d) GFK (68.34%), (e) EMAP (78.69%), (f) LBP (63.43%), (g) SPL (56.33%), (h) TwoCnn (65.30%), (i) CNN (46.09%), (j) ResNet (26.61%), and (k) GF-3DDA (81.86%).

VI. Conclusion

In this article, we present a GF-3DDA approach for HSI classification. At first, according to the joint spatial–spectral characteristics of HSI, traditional 2-D SG and 2-D DtG operators are extended to 3-D forms, which are, respectively, called 3DSG and 3DDtG. Through incorporating the 3-D Gabor transform, the so-called 3DSGG and 3DDGG features can be thus extracted. In the following, the CORAL domain adaptation method is improved by integrating the fourth-order statistics with the second-order one, and the statistics fusion-based CORAL method is sequentially applied on the three directions of HSI to boost the discriminative ability of the achieved features. Finally, ELM classifier-based multitask learning is utilized to perform pixel classification on a number of domain-adapted feature data.

Experimental results of a number of methods on four real separated HSI data sets constantly demonstrate the superiority and stability of the proposed GF-3DDA approach. Especially, three parts of GF-3DDA, including the 3DSGG and 3DDGG filters, 3DBCORAL-SF domain adaptation, and ELM-based multitask learning, are organically integrated, which could make full use of the internal properties of HSI. Finally, it is worth pointing out that the transfer learning scheme provides a feasible way to solve the challenging small sample set problem of HSI classification, which should be paid more attention in future research.

References

[1] P. W. Yuen and M. Richardson, "An introduction to hyperspectral imaging and its application for security, surveillance and target acquisition," *Imag. Sci. J.*, vol. 58, no. 5, pp. 241–253, Oct. 2010.

[2] A. Plaza *et al.*, "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, no. 1, pp. 110–122, Sep. 2009.

[3] D. Manolakis, D. Mardon, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–115, 2003.

[4] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[5] F. Wang, R. Zhang, and Q. Wu, "Hyperspectral image classification based on PCA network," in *Proc. 8th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–4.

[6] M. Imani and H. Ghassemian, "Band clustering-based feature extraction for classification of hyperspectral images using limited training samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 8, pp. 1325–1329, Aug. 2014.

[7] C. Yu, Y. Wang, M. Song, and C.-I. Chang, "Class signature-constrained background-suppressed approach to band selection for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 14–31, Jan. 2019.

[8] P. Su, D. Liu, X. Li, and Z. Liu, "A saliency-based band selection approach for hyperspectral imagery inspired by scale selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 572–576, Apr. 2018.

[9] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016.

[10] C. Wang, M. Gong, M. Zhang, and Y. Chan, "Unsupervised hyperspectral image band selection via column subset selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1411–1415, Jul. 2015.

[11] X. Kang, S. Li, and J. A. Benediktsson, "Feature extraction of hyperspectral images with image fusion and recursive filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3742–3752, Jun. 2014.

[12] X. Kang, S. Li, L. Fang, and J. A. Benediktsson, "Intrinsic image decomposition for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2241–2253, Apr. 2015.

[13] J. Lin, C. He, Z. J. Wang, and S. Li, "Structure preserving transfer learning for unsupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1656–1660, Oct. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:21:45 UTC from IEEE Xplore. Restrictions apply.

[14] Y. Qin, L. Bruzzone, and B. Li, "Tensor alignment based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9290–9307, Nov. 2019.

[15] X. Wang, Y. Li, and Y. Cheng, "Hyperspectral image classification based on unsupervised heterogeneous domain adaptation CycleGan," *Chin. J. Electron.*, vol. 29, no. 4, pp. 608–614, Jul. 2020.

[16] F. Liu, G. Zhang, and J. Lu, "Heterogeneous domain adaptation: An unsupervised approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 12, pp. 5588–5602, Dec. 2020.

[17] H. Zuo, J. Lu, G. Zhang, and W. Pedrycz, "Fuzzy rule-based domain adaptation in homogeneous and heterogeneous spaces," *IEEE Trans. Fuzzy Syst.*, vol. 27, no. 2, pp. 348–361, Feb. 2019.

[18] F. Liu, G. Zhang, and J. Lu, "Multi-source heterogeneous unsupervised domain adaptation via fuzzy-relation neural networks," *IEEE Trans. Fuzzy Syst.*, early access, Aug. 20, 2020, doi: 10.1109/TFUZZ.2020.3018191.

[19] C. Shen and Y. Guo, "Unsupervised heterogeneous domain adaptation with sparse feature transformation," in *Proc. Asian Conf. Mach. Learn.*, 2018, pp. 375–390.

[20] Y.-H.-H. Tsai, Y.-R. Yeh, and Y.-C.-F. Wang, "Learning cross-domain landmarks for heterogeneous domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 5081–5090.

[21] Q. Shi, Y. Zhang, X. Liu, and K. Zhao, "Regularised transfer learning for hyperspectral image classification," *IET Comput. Vis.*, vol. 13, no. 2, pp. 188–193, Mar. 2019.

[22] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, "A theory of learning from different domains," *Mach. Learn.*, vol. 79, nos. 1–2, pp. 151–175, May 2010.

[23] L. Yan, B. Fan, S. Xiang, and C. Pan, "Adversarial domain adaptation with a domain similarity discriminator for semantic segmentation of urban areas," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 1583–1587.

[24] Z. Wang, B. Du, Q. Shi, and W. Tu, "Domain adaptation with discriminative distribution and manifold embedding for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1155–1159, Jul. 2019.

[25] S. Sankaranarayanan, Y. Balaji, C. D. Castillo, and R. Chellappa, "Generate to adapt: Aligning domains using generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8503–8512.

[26] X. Zhou and S. Prasad, "Domain adaptation for robust classification of disparate hyperspectral images," *IEEE Trans. Comput. Imag.*, vol. 3, no. 4, pp. 822–836, Dec. 2017.

[27] P. Morerio and V. Murino, "Correlation alignment by Riemannian metric for domain adaptation," May 2017, *arXiv:1705.08180*. [Online]. Available: http://arxiv.org/abs/1705.08180

[28] L. Zhang and D. Zhang, "Robust visual knowledge transfer via extreme learning machine-based domain adaptation," *IEEE Trans. Image Process.*, vol. 25, no. 10, pp. 4959–4973, Oct. 2016.

[29] A. J. Ma, P. C. Yuen, and J. Li, "Domain transfer support vector ranking for person re-identification without target camera label information," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 3567–3574.

[30] A. J. Ma, J. Li, P. C. Yuen, and P. Li, "Cross-domain person reidentification using domain adaptation ranking SVMs," *IEEE Trans. Image Process.*, vol. 24, no. 5, pp. 1599–1613, May 2015.

[31] M. Xu, S. Wu, X.-Y. Jing, and J. Yang, "Kernel subspace alignment for unsupervised domain adaptation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 2880–2884.

[32] Y.-R. Yeh, C.-H. Huang, and Y.-C.-F. Wang, "Heterogeneous domain adaptation and classification by exploiting the correlation subspace," *IEEE Trans. Image Process.*, vol. 23, no. 5, pp. 2009–2018, May 2014.

[33] R. Gopalan, R. Li, and R. Chellappa, "Unsupervised adaptation across domain shifts by generating intermediate data representations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 11, pp. 2288–2302, Nov. 2014.

[34] B. Sun, J. Feng, and K. Saenko, "Correlation alignment for unsupervised domain adaptation," in *Domain Adaptation in Computer Vision Applications*. Cham, Switzerland: Springer, 2017, pp. 153–171.

[35] M. Ye, Y. Qian, J. Zhou, and Y. Y. Tang, "Dictionary learning-based feature-level domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1544–1562, Mar. 2017.

[36] R. Remus, "Domain adaptation using domain similarity- and domain complexity-based instance selection for cross-domain sentiment analysis," in *Proc. IEEE 12th Int. Conf. Data Mining Workshops (ICDMW)*, Dec. 2012, pp. 717–723.

[37] W. Pengfei, K. Yiping, and G. C. Keong, "Domain specific feature transfer for hybrid domain adaptation," in *Proc. IEEE ICDM*, Nov. 2017, pp. 1027–1032.

[38] B. Sun, X. Kang, S. Li, and J. A. Benediktsson, "Random-Walker-based collaborative learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 212–222, Jan. 2017.

[39] S. Jia, L. Shen, and Q. Li, "Gabor feature-based collaborative representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1118–1129, Feb. 2015.

[40] S. Zhang, T. Wakahara, and Y. Yamashita, "Image matching using GPT correlation associated with simplified hog patterns," in *Proc. IEEE IPTA*, Nov. 2017, pp. 1–6.

[41] L. D. Griffin, "The atlas structure of images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 1, pp. 234–245, Jan. 2019.

[42] T. R. Reed, "The representation and coding of volumetric images using the 3-D derivative of Gaussian transform," in *Proc. Int. Conf. Image Process.*, 2001, pp. 585–588.

[43] P. Koniusz, Y. Tas, and F. Porikli, "Domain adaptation by mixture of alignments of second or higher-order scatter tensors," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7139–7148.

[44] G.-B. Huang, Q.-Y. Zhu, and C.-K. Siew, "Extreme learning machine: Theory and applications," *Neurocomputing*, vol. 70, nos. 1–3, pp. 489–501, Dec. 2006.

[45] J. Xia, M. D. Mura, J. Chanussot, P. Du, and X. He, "Random subspace ensembles for hyperspectral image classification with extended morphological attribute profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 4768–4786, Sep. 2015.

[46] S. Jia, B. Deng, J. Zhu, X. Jia, and Q. Li, "Local binary pattern-based hyperspectral image classification with superpixel guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 749–759, Feb. 2018.

[47] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011.

[48] B. Gong, Y. Shi, F. Sha, and K. Grauman, "Geodesic flow kernel for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2066–2073.

[49] Q. Wang and T. Breckon, "Unsupervised domain adaptation via structured prediction based selective pseudo-labeling," in *Proc. AAAI*, 2020, vol. 34, no. 4, pp. 6243–6250.

[50] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[51] J. Yang, Y. Zhao, J. C.-W. Chan, and C. Yi, "Hyperspectral image classification using two-channel deep convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5079–5082.

[52] Y. Jiang, Y. Li, and H. Zhang, "Hyperspectral image classification based on 3-D separable ResNet and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1949–1953, Dec. 2019.

[53] S. Israni and S. Jain, "Edge detection of license plate using sobel operator," in *Proc. ICEEOT*, 2016, pp. 3561–3563.

[54] N. Kanopoulos, N. Vasanthavada, and R. L. Baker, "Design of an image edge detection filter using the sobel operator," *IEEE J. Solid-State Circuits*, vol. SSC-23, no. 2, pp. 358–367, Apr. 1988.

[55] L. Shen and S. Jia, "Three-dimensional Gabor wavelets for pixel-based hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 5039–5046, Dec. 2011.

[56] S. Jia, L. Shen, J. Zhu, and Q. Li, "A 3-D Gabor phase-based coding and matching framework for hyperspectral imagery classification," *IEEE Trans. Cybern.*, vol. 48, no. 4, pp. 1176–1188, Apr. 2018.

[57] A. Samat, P. Du, S. Liu, J. Li, and L. Cheng, "E^2LMs: Ensemble extreme learning machines for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1060–1069, Apr. 2014.

[58] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[59] L. Ma, M. M. Crawford, L. Zhu, and Y. Liu, "Centroid and covariance alignment-based domain adaptation for unsupervised classification of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2305–2323, Apr. 2019.

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